THE CLASS OF A FINITELY SUPPORTED SPECTRAL SEQUENCE IN K-ZERO EQUALS THE CLASS OF ITS COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the closing sentence 'and the result is clear' follows a displayed finite filtration whose graded pieces were identified in the cited convergence lemma, so it closes the argument rather than deferring a step.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Filtered complexes and the spectral sequence attached to one. Same chapter, Definition definition-filtered together with the construction of the spectral sequence of a filtered complex. Supplies the filtered complex, the finiteness of the filtration and the pages of the spectral sequence on which the argument operates.
- Convergence of the spectral sequence of a finitely filtered complex. Same chapter, Lemma lemma-biregular-ss-converges. Supplies that each cohomology object has a finite filtration whose graded pieces are the limiting terms, which is what turns the alternating sums into the displayed identity.
- The zeroth K-group of an abelian category. Same chapter, Definition definition-K-zero. Supplies the group in which the displayed identity is taken and in which alternating sums of classes are computed.

Source reference. Original definition 0121: sources/stacks-homology/pinned-original-source.tex, source lines 4884–4884. Navigation pointer invoked at proof line 6582; the counted span contains the pointer only and the cited result is not reproduced

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Source reference. Original lemma 012W: sources/stacks-homology/pinned-original-source.tex, source lines 6579–6579. Navigation pointer invoked at proof line 6666; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. Assume that the filtration on each K^n is finite (see Definition 0121) and that for some r we have only a finite number of nonzero $E_r^{p,q}$. Then only a finite number of $H^n(K^\bullet)$ are nonzero and we have*

$$\sum (-1)^n [H^n(K^\bullet)] = \sum (-1)^{p+q} [E_r^{p,q}]$$

in $K_0(\mathcal{A}')$ where $\mathcal{A}'$ is the smallest weak Serre subcategory of $\mathcal{A}$ containing the objects $E_r^{p,q}$.

Proof. Denote E_r^{even} and E_r^{odd} the even and odd part of E_r defined as the direct sum of the (p, q) components with $p + q$ even and odd. The differential d_r defines maps $\varphi : E_r^{even} \to E_r^{odd}$ and $\psi : E_r^{odd} \to E_r^{even}$ whose compositions either way give zero. Then we see that

$$\begin{aligned} [E_r^{even}] - [E_r^{odd}] &= [\mathrm{Ker}(\varphi)] + [\mathrm{Im}(\varphi)] - [\mathrm{Ker}(\psi)] - [\mathrm{Im}(\psi)] \\ &= [\mathrm{Ker}(\varphi)/\mathrm{Im}(\psi)] - [\mathrm{Ker}(\psi)/\mathrm{Im}(\varphi)] \\ &= [E_{r+1}^{even}] - [E_{r+1}^{odd}] \end{aligned}$$

Note that all the intervening objects are in the smallest Serre subcategory containing the objects $E_r^{p,q}$. Continuing in this manner we see that we can increase r at will. Since there are only a finite number of pairs (p, q) for which $E_r^{p,q}$ is nonzero, a property which is inherited by $E_{r+1}, E_{r+2}, \ldots$, we see that we may assume that $d_r = 0$. At this stage we see that $H^n(K^\bullet)$ has a finite filtration (Lemma 012W) whose graded pieces are exactly the $E_r^{p,n-p}$ and the result is clear. □